



Predicting Child Birth Weights in Bangladesh

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I. Project Motivation

II. Exploratory Data Analysis

III. Methodology

IV. Results

V. Conclusions



Background

- We analyzed a particular dataset from a large survey program known as the Demographic Health Surveys (DHS)
- DHS surveys developing countries on multiple aspects of health
- We analyzed the set of surveys related to birth health for Bangladesh
- Focused on birth weight for the children surveyed from 2013-2018



Project Motivation

- Globally, 1 in 7 babies is born with low birth weight (<2,500 grams)
- It increases the risk of:
 - Neonatal mortality
 - Developmental delays
 - Chronic diseases later in life (e.g., diabetes, heart disease)
- Disproportionately affects low-resource settings, making it a priority for global health equity



Project Motivation

“Low birth weight is a significant predictor of newborn health and survival. It is a key indicator of maternal health, nutrition, and care.”

- World Health Organization (WHO)



So why model it?

Being able to correctly model birth weight would help:

- Enable early identification of at-risk pregnancies
- Support targeted interventions during pregnancy
- Aid resource allocation in maternal and child health programs
- Guide data-driven public health policies



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Research Question:

Can we identify regions of Bangladesh that are particularly susceptible to negative birth outcomes?



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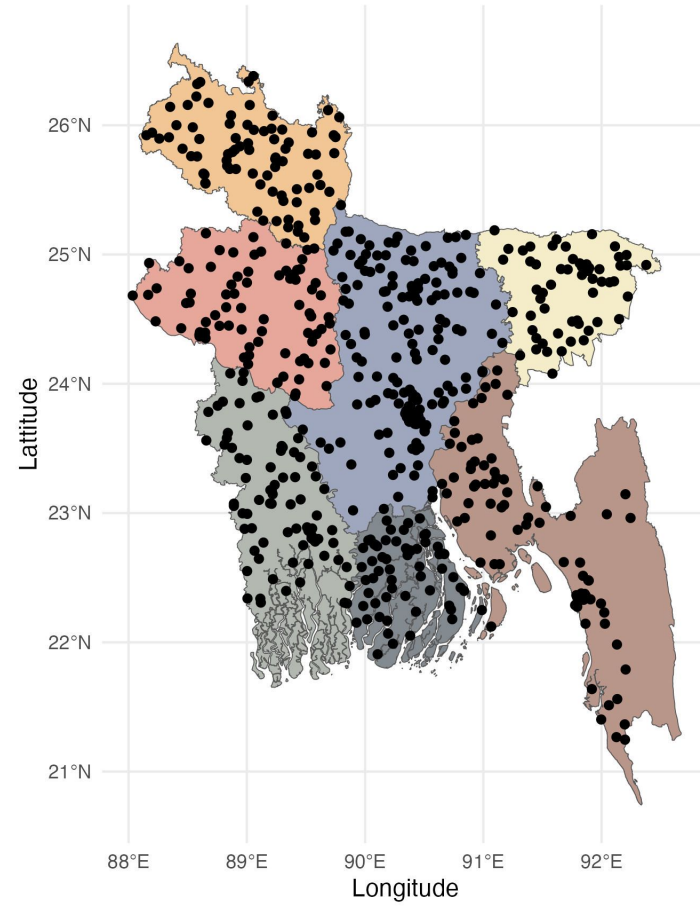


Dataset Analysis

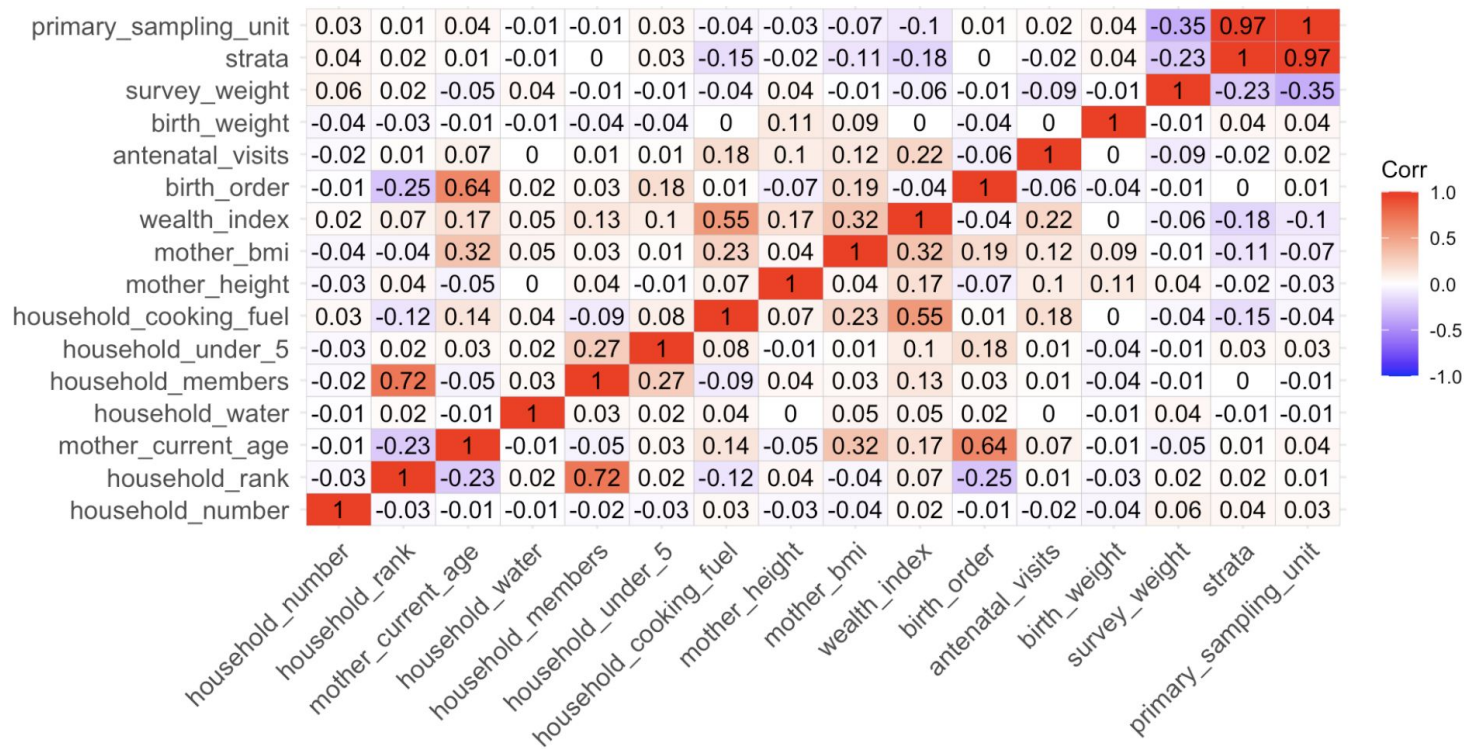
- The dataset contains 2383 observations
 - Feasibility of fitting a full GP becomes questionable
- 29 covariates for each observation
 - 4 broad categories: Household and Demographic, Maternal, Child, Household Infrastructure
 - Mix of categorical, numeric, and ordinal data

Very common in social sciences do an analysis with covariates not included within the data to understand anthropogenic effects on health.

Map of DHS Clusters



Covariate Analysis - Pairwise Correlation



Covariate Selection



- Multicollinearity doesn't carry the same risks when using GPs compared to simple Linear Regression however it is useful in identifying redundant covariates
- The motivation behind our selection process was to maintain a simple model with few covariates and select ones which appear frequently in existing studies so that a literature review would be possible with our results
- Tao, J., Yuan, Z., Sun, L. et al. found that both the maternal BMI was the most significant predictor of birth weight
- Classification targets are often stratified across gestational age and sex therefore we decided to include the child's sex as a covariate
- Existing literature largely focuses on maternal/paternal physical characteristics therefore we wanted to introduce a socio-economic factor as well by including the wealth index as a covariate



Model Parameters

Covariates

- Mother BMI
- Sex of Child
- Wealth Index

Target

- Birth Weight



Data Preprocessing

- Wealth index, being an ordinal feature, is treated as a numeric thus there is an assumption that the wealth index is linearly associated with household wealth
- To create pseudo-unique locations, small i.i.d. Gaussian noise ($SD = 0.001$ degrees) was added to latitude/longitude
- Observations are spatially indexed using these randomized coordinates



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A brief overview of the methods

Covariance Function: We chose the exponential covariance function

$$C_{\sigma^2, \phi}(s, s') = \sigma^2 \exp\{-\phi \|s - s'\|\}$$

- Simple and parsimonious, easy to interpret
- Only require two parameter to estimate
- Assuming the spatial process is stationary and isotropic

Method:

- Full Gaussian Process with Universal Kriging (Full GP)
- Vecchia Approximation
- Nearest Neighbor Gaussian Process (NNGP)
- Meshed Gaussian Process

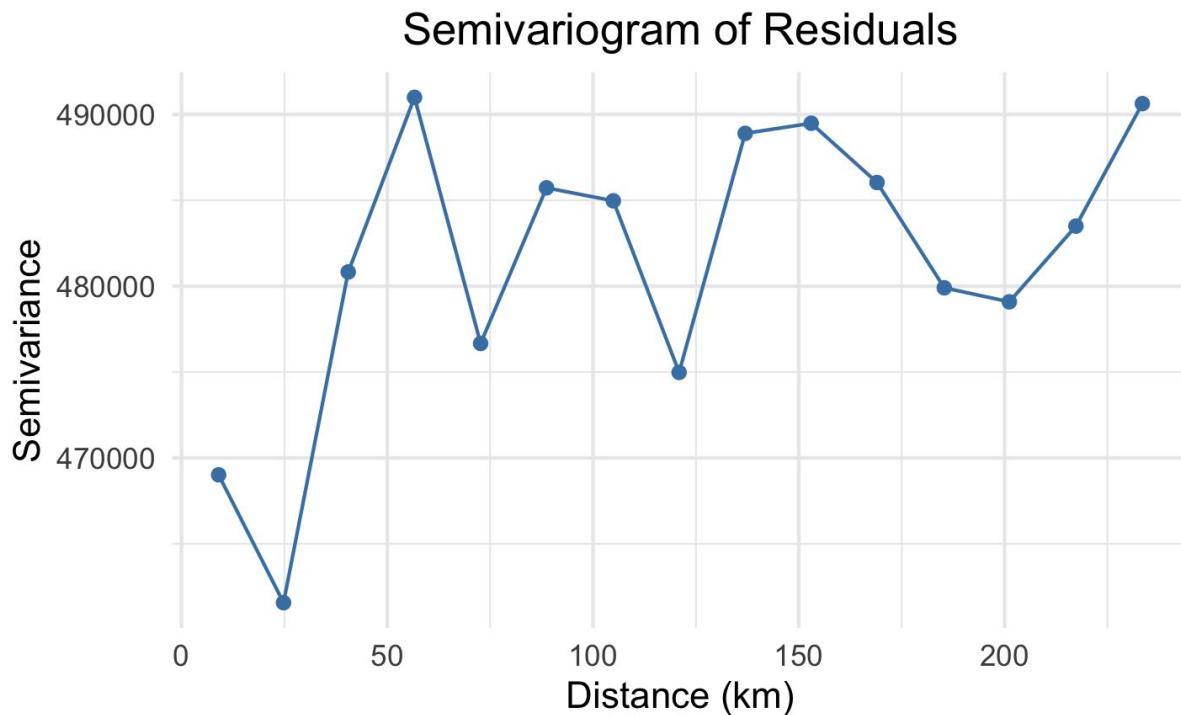


Full GP Regression with Universal Kriging

$$Y(s) = X(s)^T \beta + w(s) + \epsilon$$

- Two stage fitting:
 - 1. Get the ordinary least squares estimate for Beta
 - 2. Estimate covariance parameters from semivariogram of the residuals
- Implemented with “gstat” library

Full GP Regression with Universal Kriging



Full GP Regression with Universal Kriging



Utilizing the fact that the semivariogram and the covariance function are related.

- Semivariogram, $\gamma(h)$
- Covariance function, $C(d)$
- Then, $\gamma(h) = C(0) - C(h)$

Therefore for the exponential covariance,

$$\gamma(h) = \sigma^2 \left(1 - \exp \left(-\frac{h}{\phi} \right) \right)$$

Full GP Regression with Universal Kriging



Since we want to include a nugget effect, we include it:

$$\gamma(h) = \tau^2 + \sigma^2 \left(1 - \exp \left(-\frac{h}{\phi} \right) \right)$$

Full GP Regression with Universal Kriging



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Full GP Regression with Universal Kriging

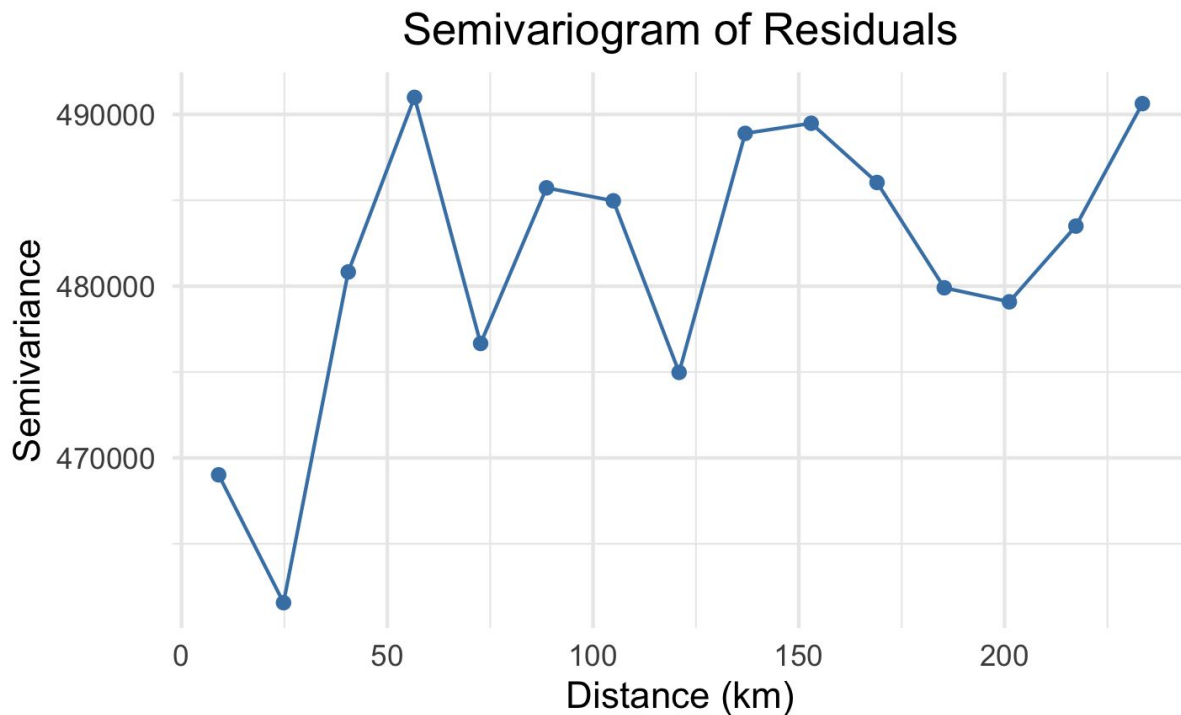
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$$\gamma(h) = \tau^2 + \sigma^2 \left(1 - \exp \left(-\frac{h}{\phi} \right) \right)$$

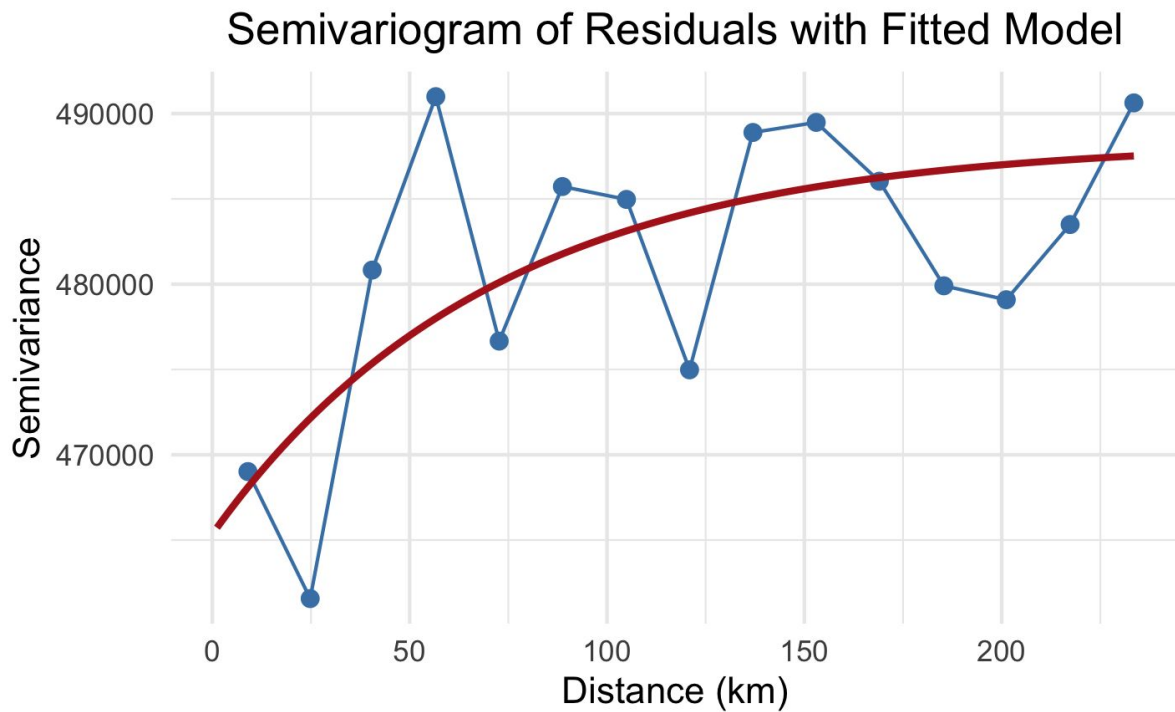
This is done by minimizing the sum of squared differences:

$$\min_{\tau^2, \sigma^2, \phi} \sum_{k=1}^K w_k \left[\hat{\gamma}(h_k) - \left(\tau^2 + \sigma^2 \left(1 - \exp \left(-\frac{h_k}{\phi} \right) \right) \right) \right]^2$$

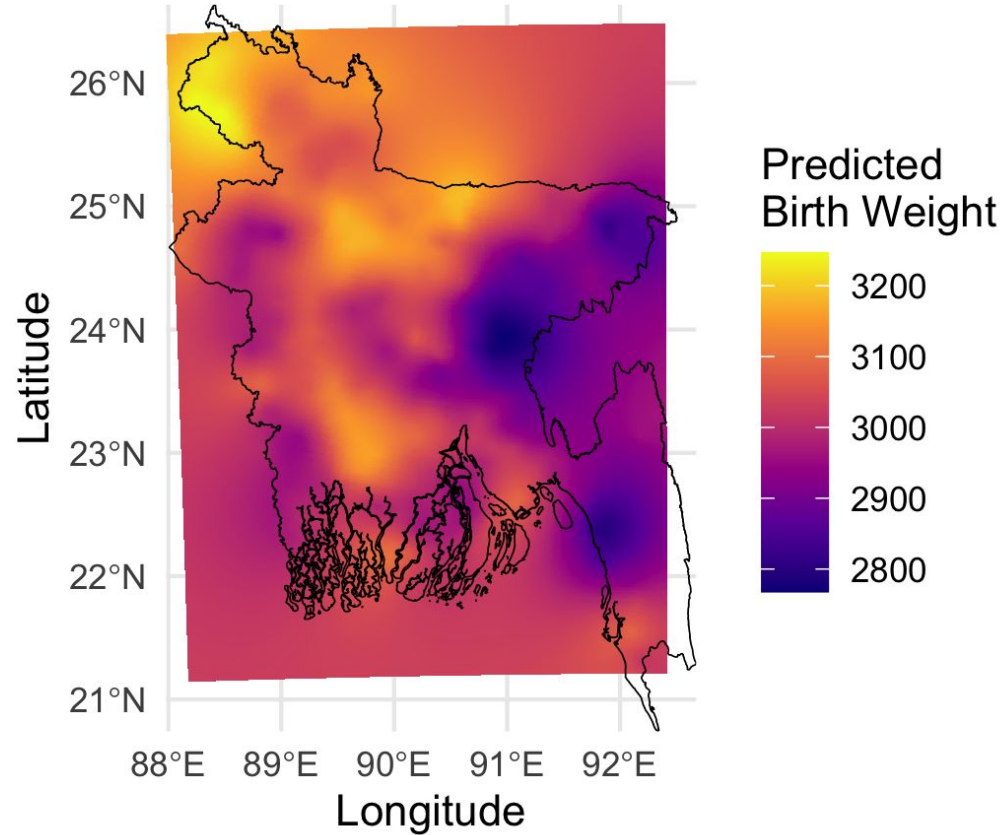
Full GP Regression with Universal Kriging



Full GP Regression with Universal Kriging



Universal Kriging Birth Weight Predictions



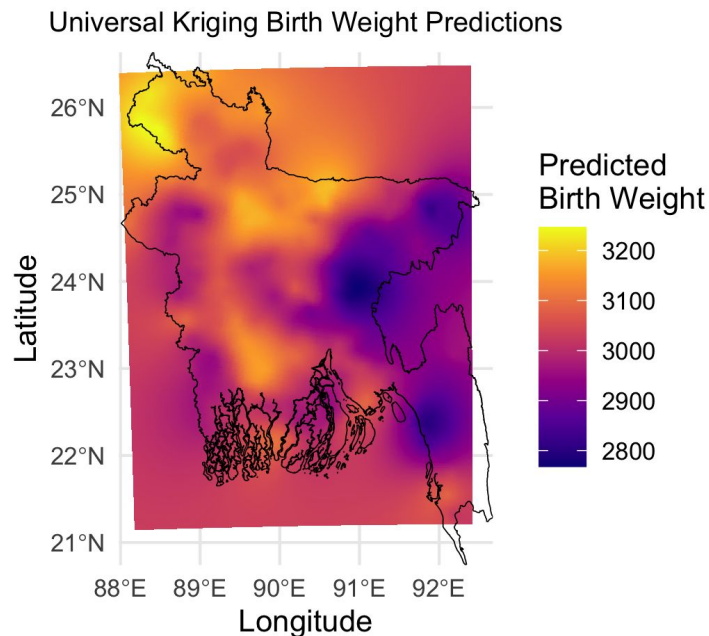
Full GP Regression with Universal Kriging

Results:

Eastern and south eastern coast of Bangladesh appears to have lower predicted birth weights than other parts of the country when controlling for maternal health and wealth.

Limitations:

Estimated linear regression and Gaussian process covariance structure independently.



Vecchia Approximation



Turnaround time to fit, evaluate and calibrate the priors of a full GP **do not fit project constraints**

Possible Solution:

- Vecchia approximations offer consistent and efficient estimation of covariance and trend parameters in the GP model and allows principled fitting of universal kriging models within the approximate GP framework

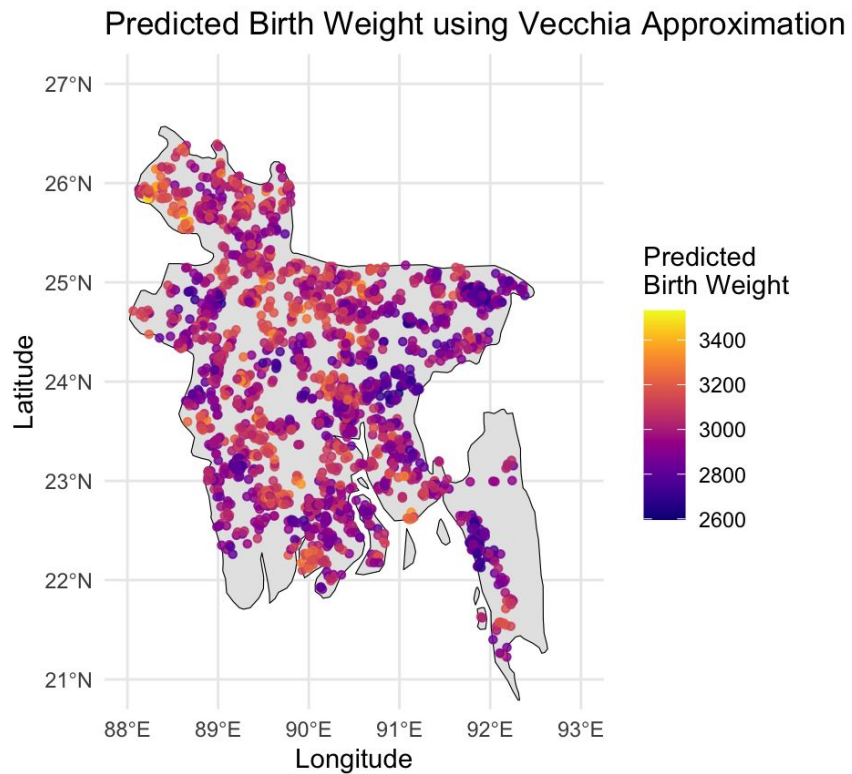
Benefits:

- Maximum Marginal Likelihood (MML) eliminates the need to specify the prior distribution of the covariance function parameters, avoiding reliance on cross-validation or empirical variogram fitting
- MML-fitted Vecchia models can serve as initialization or proposal mechanisms for fully Bayesian NNGP estimation, improving mixing and convergence

Limitations:

- The MML objective is sensitive to the choice of conditioning sets; poor conditioning orders can degrade parameter estimation quality
- Approximation error in the marginal likelihood can propagate into bias in estimated parameters

Vecchia Approximation





Nearest Neighbor Gaussian Process

$$Y \sim NNGP(X\beta, C_{\sigma^2, \phi}(\cdot, \cdot))$$

$$\beta \sim N(0, V_\beta)$$

$$C_{\sigma^2, \phi}(s, s') = \sigma^2 \exp\{-\phi\|s - s'\|\} + \tau^2 \mathbf{1}_{s=s'}$$

$$\phi \sim U(5, 10)$$

$$\sigma^2 \sim \text{InvGamma}(2, 2)$$

$$\tau^2 \sim \text{InvGamma}(2, 0.01)$$

- For spatial decay parameter $\phi \sim U(5, 10)$, we leverage prior knowledge from the Vecchia approximated GP marginalized likelihood, while σ^2 and τ^2 are non-informative inverse-gamma priors to regularize estimation without overly constraining the posterior.
- We set the number of nearest neighbors to **10** to balance approximation accuracy and computational efficiency.



Posterior Sampling for NNGP

Sampling Regression Coefficients β :

Given y, τ^2, w , the conditional posterior distribution of β is multivariate normal. Sampling is straightforward via Gibbs sampling.

Sampling Random Effects w :

Given β, y , and ϕ, σ^2, τ^2 , the conditional posterior distribution of w is multivariate normal. Updated via Gibbs sampling, using sparse precision matrix from NNGP prior for computational efficiency.

Sampling τ^2

Given β, w, y , the conditional posterior of τ^2 follows inverse Gamma distribution, which allows direct Gibbs sampling.

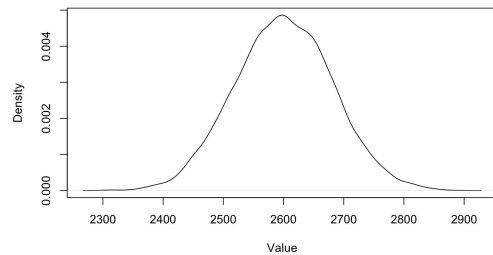
Sampling ϕ, σ^2 :

Employ Metropolis-Hasting algorithm to update the spatial covariance parameter ϕ, σ^2 based on current values of w .

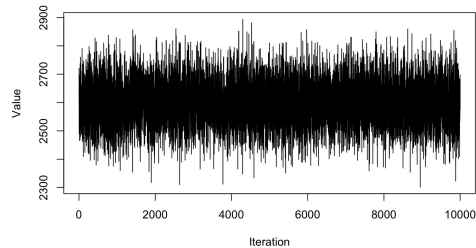


Nearest Neighbor Gaussian Process - □

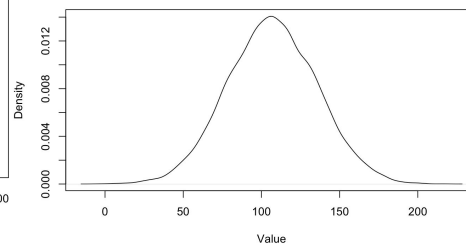
Density of Beta 1 (Intercept)



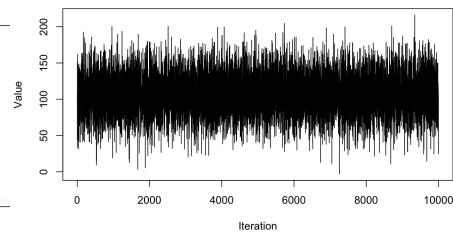
Trace of Beta 1 (Intercept)



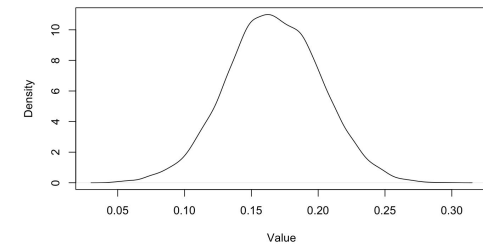
Density of Beta 3 (sex_of_childmale)



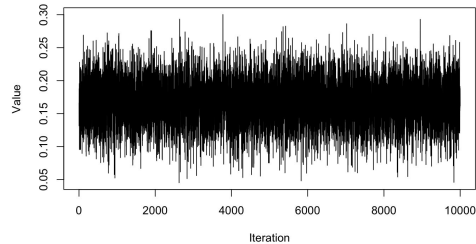
Trace of Beta 3 (sex_of_childmale)



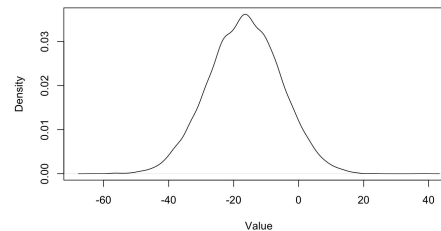
Density of Beta 2 (mother_bmi)



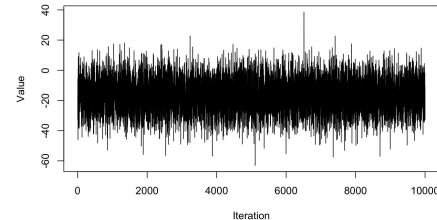
Trace of Beta 2 (mother_bmi)



Density of Beta 4 (wealth_index)

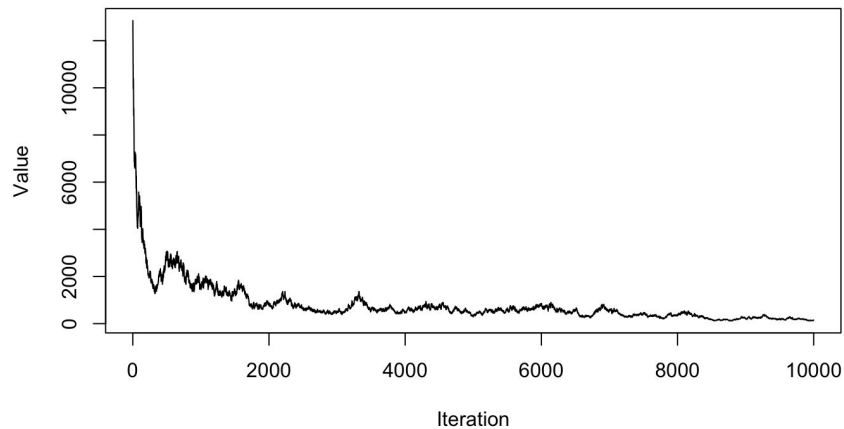


Trace of Beta 4 (wealth_index)

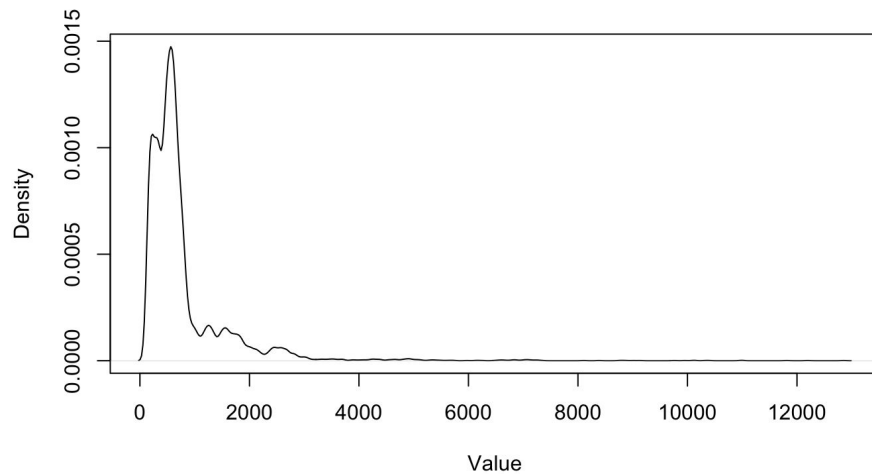


Nearest Neighbor Gaussian Process - ϕ

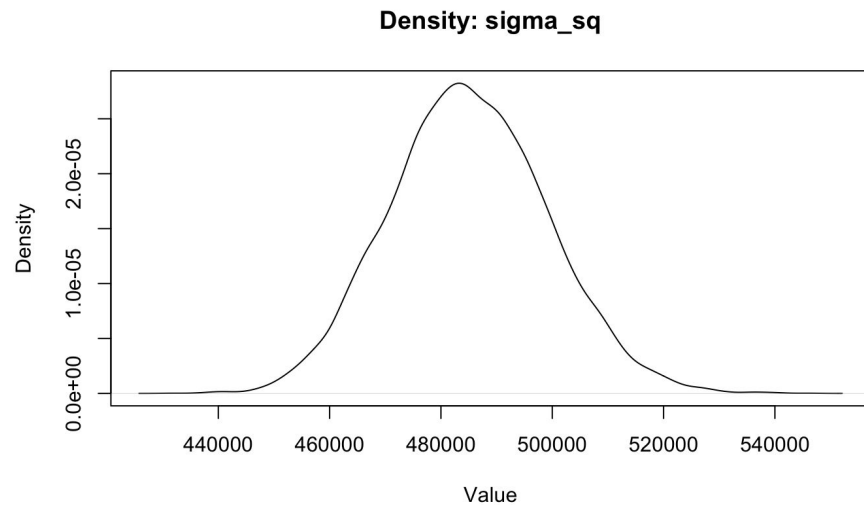
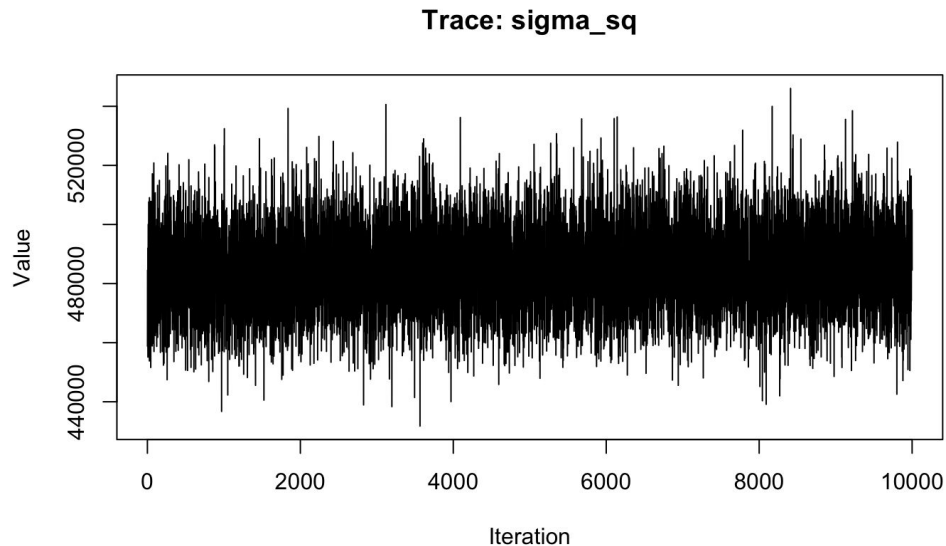
Trace: phi



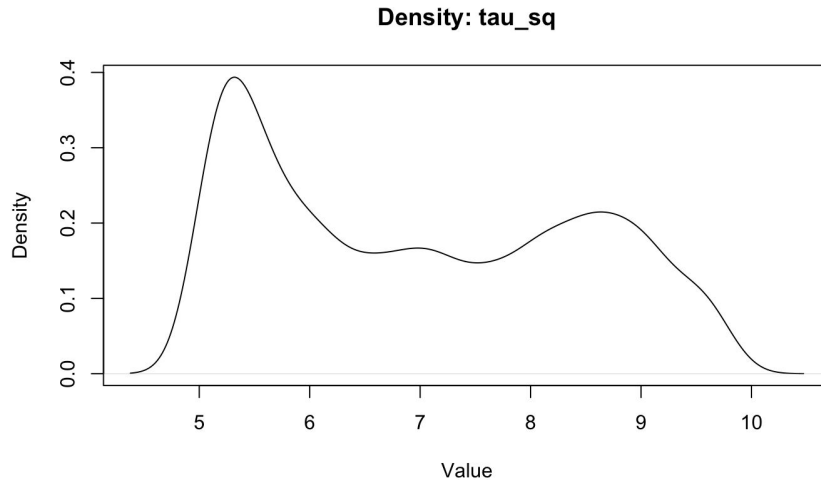
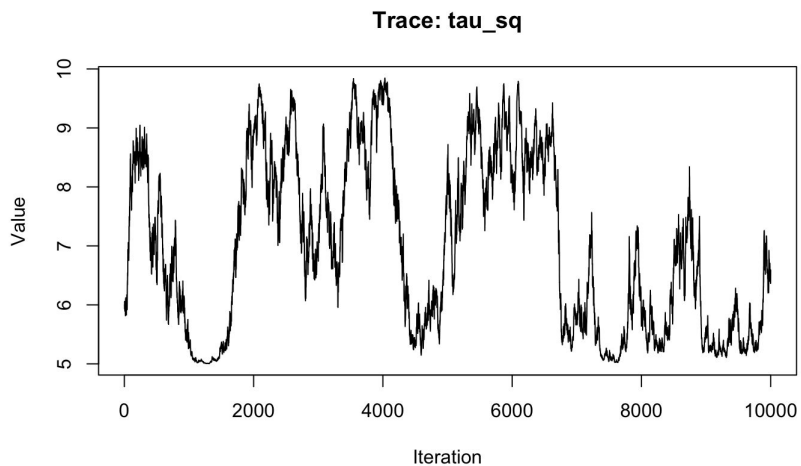
Density: phi



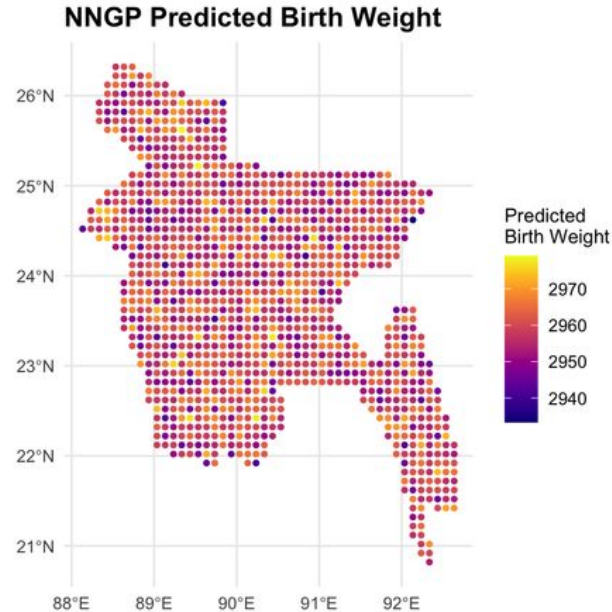
Nearest Neighbor Gaussian Process - σ^2



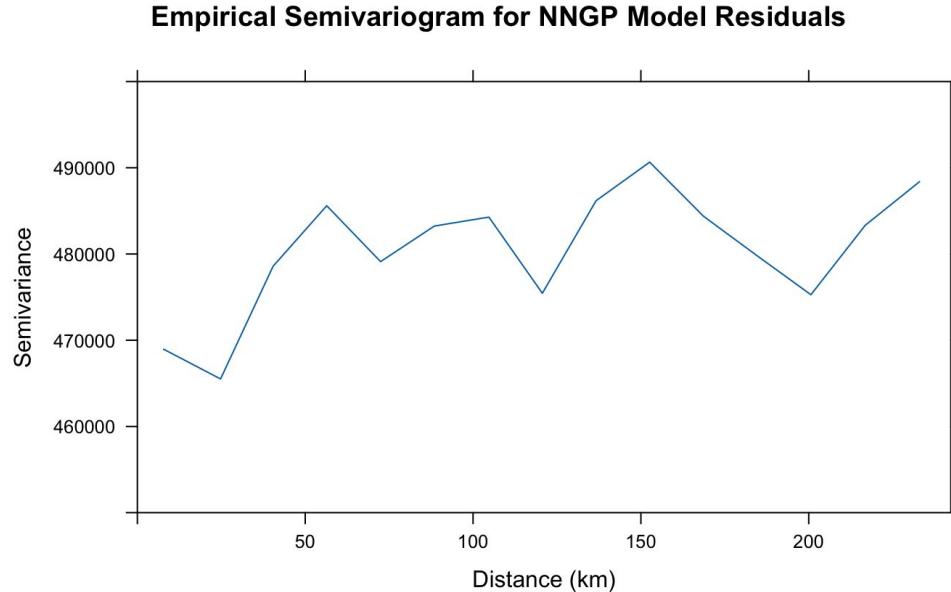
Nearest Neighbor Gaussian Process - τ^2



Nearest Neighbor Gaussian Process - Prediction



Nearest Neighbor Gaussian Process - Semivariogram





Meshed Gaussian Process

- **DAG Grid Construction & Local Conditional Modeling**

A fixed spatial grid is imposed over the domain, and each grid point connects to a small set of earlier neighbors to form a directed acyclic graph. This defines a sparse dependency structure across space.

- **Model Setting Similar to NNGP: no priors to beta**

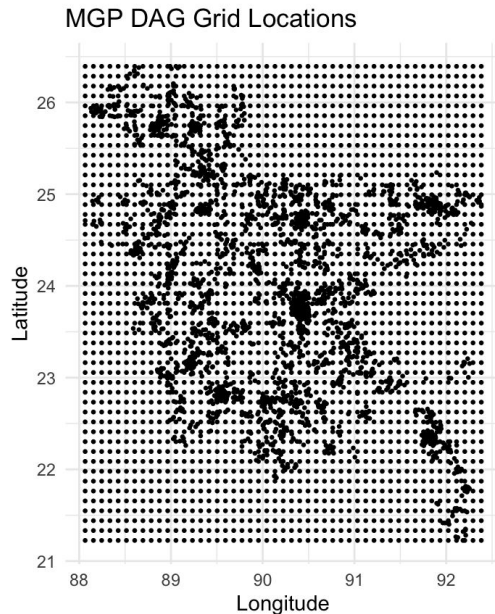
$$\tau^2 \sim \text{Inverse-Gamma}(2, 1)$$

$$\sigma^2 \sim \text{Inverse-Gamma}(2, 0.01)$$

$$\phi \sim \text{Uniform}(1, 15)$$

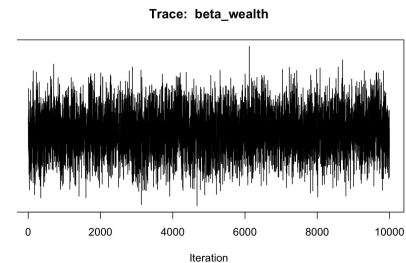
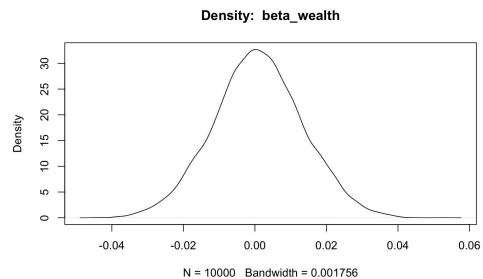
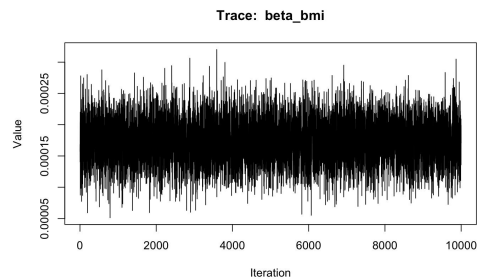
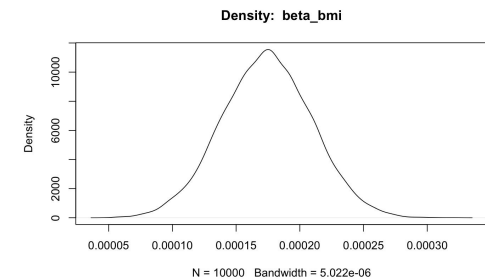
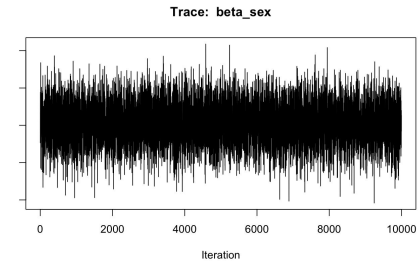
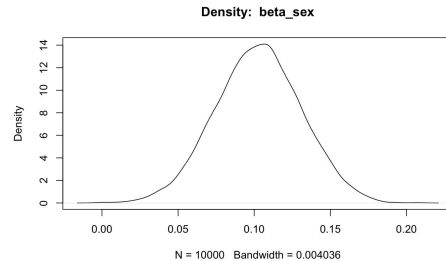
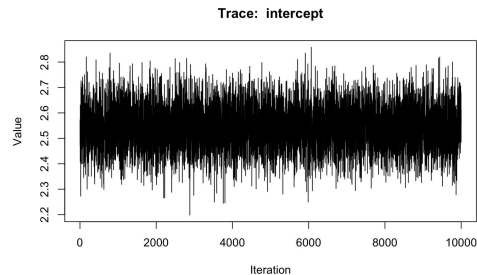
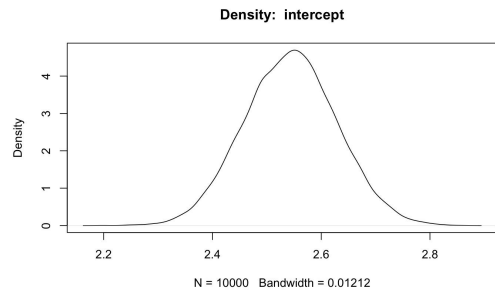
- The key benefit of the DAG structure is that it gives us scalability similar to NNGP, but with even more regularity and efficient implementation via preconstructed neighbor sets.

Meshed Gaussian Process

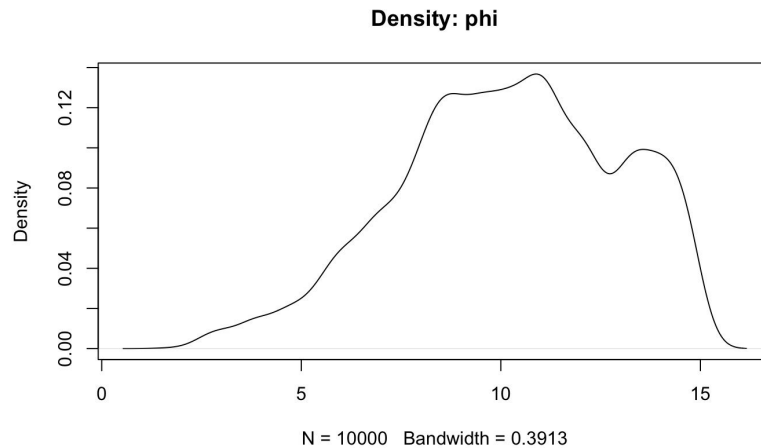
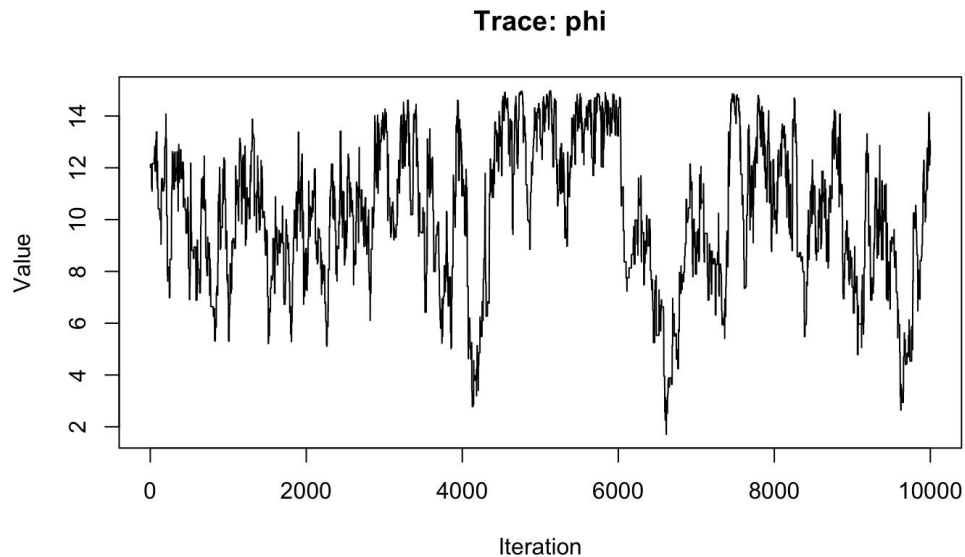


- Regularly spaced grid points are placed over the spatial domain.
- Each point connects to a few upstream neighbors to form a directed acyclic graph (DAG).
- These locations form the basis for approximating the spatial process in the Meshed GP model.
- Predictions and latent effects are computed on this grid.

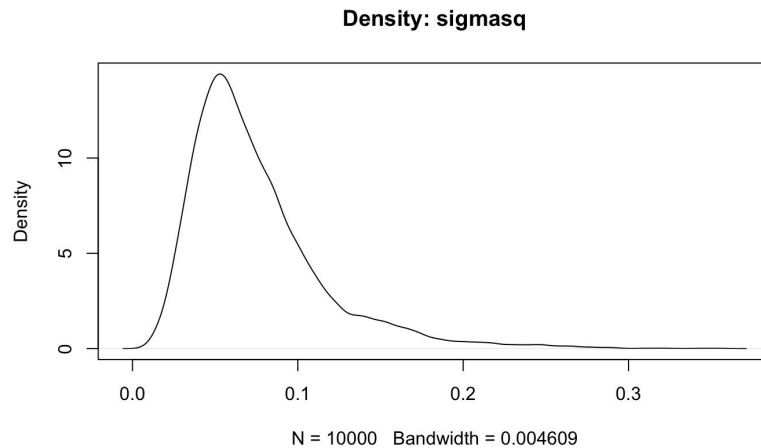
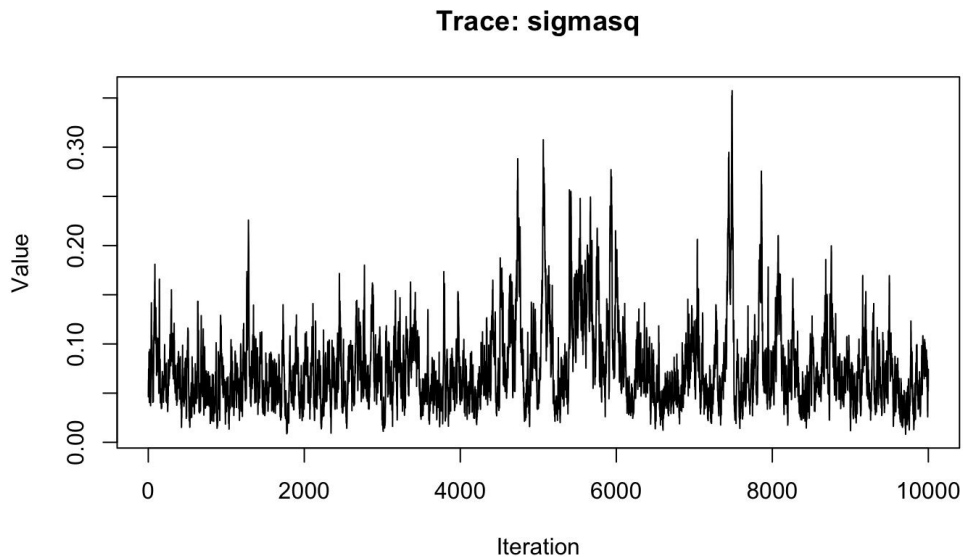
Meshed Gaussian Process - □



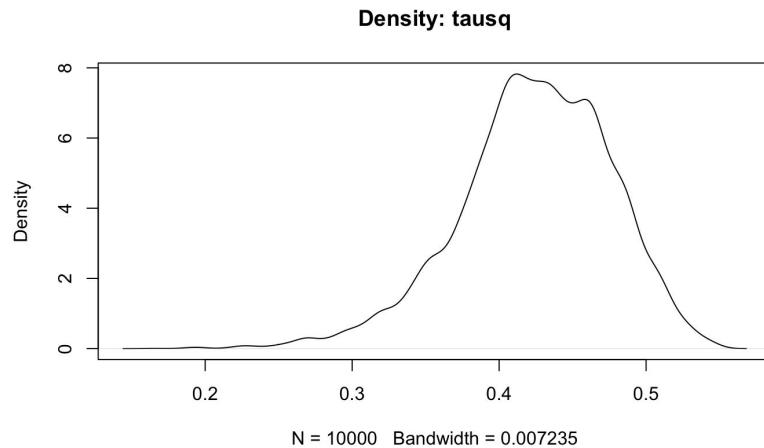
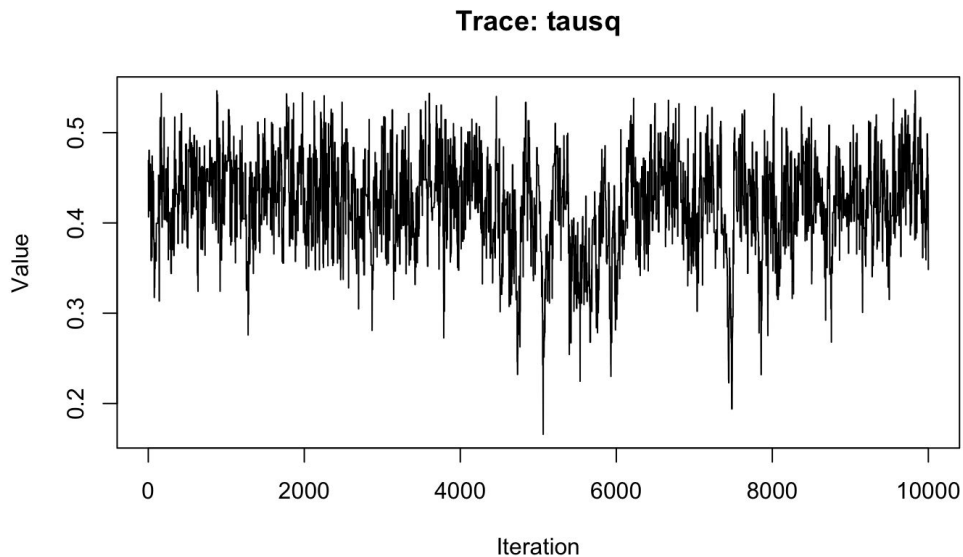
Meshed Gaussian Process - ϕ



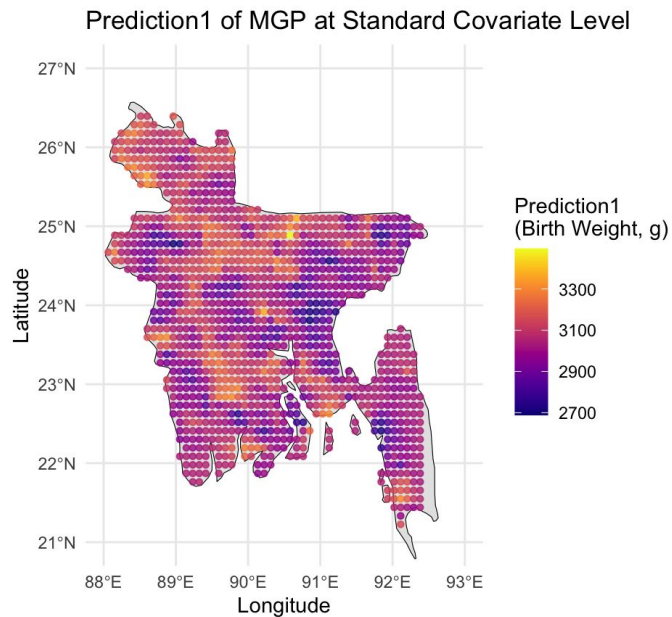
Meshed Gaussian Process - σ^2



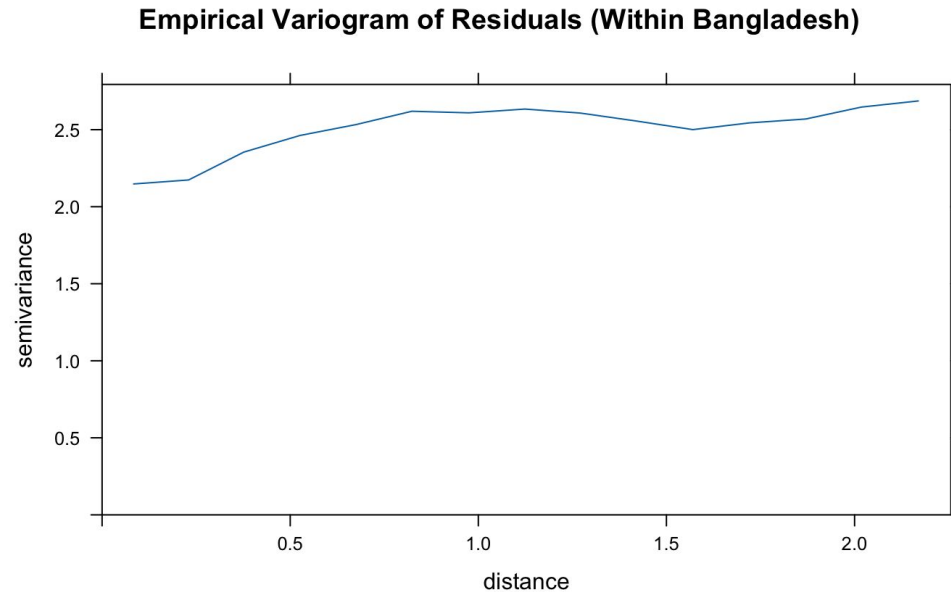
Meshed Gaussian Process - τ^2



Meshed Gaussian Process - Prediction



Meshed Gaussian Process - Residual Variogram

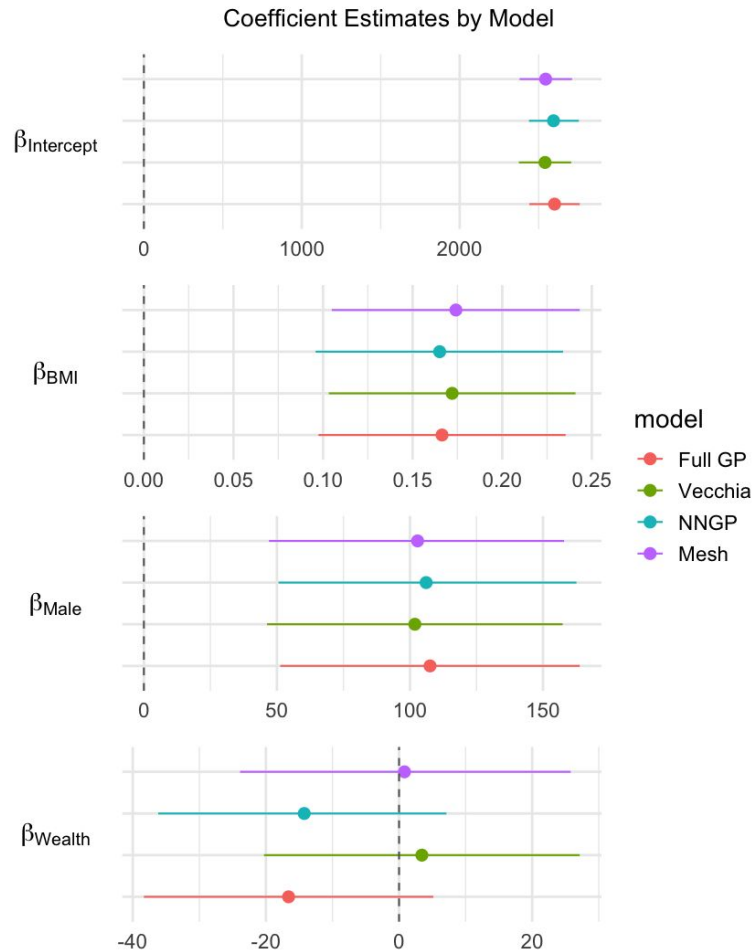




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Results

All methods produce relatively similar coefficients for the independent variables included.



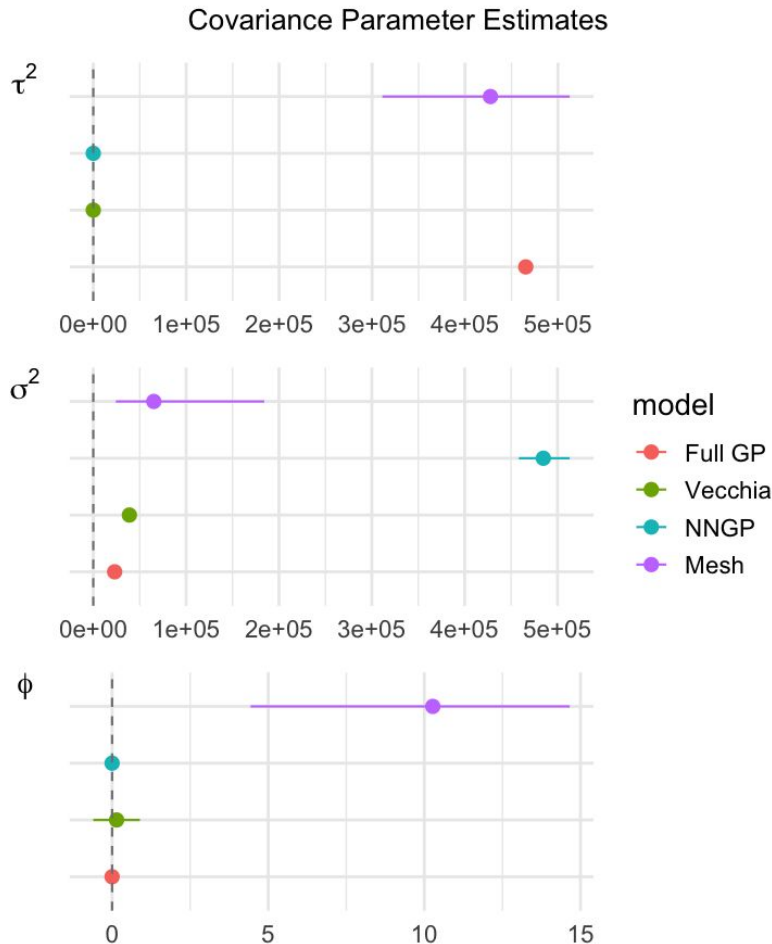
Results

For τ^2 and σ^2 :

- Difficult to differentiate between the measurement error and spatially variability
- Already low amount of spatially variability to begin

For ϕ :

- Full GP, Vecchia, and NNGP estimates the spatial correlation drops off quickly
- MeshedGP indicates the spatial correlation falls off slower but might be due to the inflated size of τ^2





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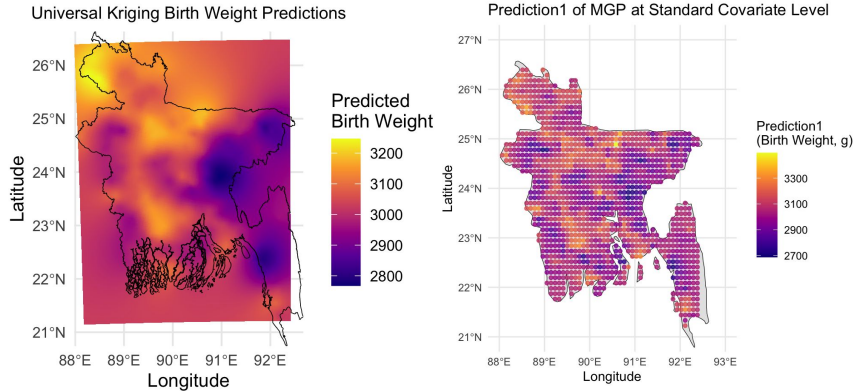
Conclusion

- NNGP provides the best tradeoff between predictive accuracy and computational efficiency. Although NNGP shows more granular predictions and captures less residual spatial variation, this might reflect the limited spatial signal in the data - an expected outcome given the low signal to noise ratio commonly observed in survey-based datasets.
- Full GP and Meshed GP produced smooth spatial prediction surfaces and captures spatial gradients - offers strong interpretability and model richness.
- The Vecchia approximation, while efficient, produced implausible fixed effects (e.g. Birth weight negatively correlated with wealth index) and may be less suitable for guiding real-world decisions.

Answer to Research Question

Q: Can we identify regions of Bangladesh that are particularly susceptible to negative birth outcomes?

A: Yes, we can identify regions of Bangladesh that are particularly susceptible to birth outcomes, especially in the eastern and coastal area of Bangladesh (seen in Full GP & meshed-GP Predictions), although some methods showed more spatial contrast than others.





Policy Relevance

The result from our spatial analysis can help identify region potentially at risk for poor birth outcomes and inform targeted maternal and child health interventions.

Stakeholders Include:

- Mothers and Children (target population)
- Policymakers allocating health resources
- Taxpayer and donor supporting evidence-based spending